

ESP32-S3 Plant Monitor — Pin Map Cheat Sheet

Rev A · project RS_ESP32S3_PlantMonitor_RevA · module ESP32-S3-WROOM-1-N8 (U1)

At a glance	I ² C: SCL = GPIO4 , SDA = GPIO5 · pull-ups R7/R8 = 4.7 kΩ to +3V3
	Sensors: BME280 = 0x76 (T/RH/pressure) · VEML7700 = 0x10 (light)
	ADC1 (attn 3, range 0–2900 mV): GPIO1 = soil (ADC1_CH0) · GPIO2 = battery (ADC1_CH1)
	Soil-probe power: GPIO21 — drive LOW = probe ON , HIGH = off (default)
	USB (native): GPIO19 = D– , GPIO20 = D+ · no separate bridge chip
	Buttons: EN = RESET (SW1) · GPIO0 = BOOT (SW2, hold low to flash)
	Strapping pins (state matters at reset): GPIO0, GPIO3, GPIO45, GPIO46

Module pin map — every used pin of U1

Pin	Name	Net	Role
2	3V3	+3V3	Module power input (C4 22 µF + C5 100 nF at the pin)
3	EN	—	Chip-enable / reset. Network: R5 10 kΩ pull-up, C6 1 µF, SW1 RESET
27	IO0	—	Boot strap: R6 10 kΩ pull-up + SW2 BOOT to GND (low = download mode)
13	IO19	USB_DN	Native USB data minus
14	IO20	USB_DP	Native USB data plus
4	IO4	I2C_SCL	I ² C clock to both sensors
5	IO5	I2C_SDA	I ² C data to both sensors
39	IO1 · ADC1_CH0	ADC_SOIL	Soil-probe analog reading (through R9/R10 divider)
38	IO2 · ADC1_CH1	BAT_SENSE	Battery voltage — half of VBAT (through R16/R17)
23	IO21	SENS_PWR_EN	Drives soil-probe power switch Q2 (LOW = probe on)
37	TXD0	— (TP12)	UART fallback transmit (recovery flashing / logs)
36	RXD0	— (TP13)	UART fallback receive
1, 40, 41	GND / pad	GND	Ground and thermal pad (pad also sinks heat)

Net glossary

Net	Meaning
USB_VBUS	Raw 5 V at the USB-C connector, before the fuse
+5V_PROT	Protected 5 V after the fuse — the 5 V rail everything uses
VSYS	System node: ~4.6 V on USB, battery voltage on battery. Feeds the regulator
+3V3	Regulated 3.3 V rail from the AP2112 (U3)
VBAT_RAW	Battery connector positive, before reverse-polarity protection
VBAT	Protected battery rail, after Q1. Charger + power path connect here
BAT_SENSE	Half of VBAT → GPIO2, for battery-percentage measurement
USB_DP / USB_DN	USB data +/–, connector to the ESP32-S3 native USB pins
I2C_SCL / I2C_SDA	Shared two-wire sensor bus: clock and data
SOIL_PWR	Switched 3.3 V that powers the soil probe only when enabled
SENS_PWR_EN	Control signal (GPIO21) for the soil-probe power switch
ADC_SOIL	Divided-down soil-probe output → GPIO1

Test points — where to put the probe

TP	Net	Check	TP	Net	Check
TP1	+5V_PROT	5 V present after the fuse	TP9	I2C_SDA	I ² C data activity
TP2	+3V3	Regulator output alive at 3.3 V	TP10	ADC_SOIL	Divided soil-probe voltage
TP3	GND	Meter / scope ground reference	TP11	VBAT	Protected battery rail; charge-voltage check
TP4	EN	Reset line high; RESET button action	TP12	TXD0	UART fallback transmit
TP5	IO0	Boot strap state; BOOT button action	TP13	RXD0	UART fallback receive
TP6	USB_DP	USB data + (after series resistor)	TP14	VSYS	System node — watch USB↔battery hand-over here
TP7	USB_DN	USB data – (after series resistor)	TP15	BAT_SENSE	Half-battery voltage into the ADC
TP8	I2C_SCL	I ² C clock activity			

Source: ESP32-S3 Plant Monitor Final Design Document, Rev A. Values are design intent — always confirm against the schematic and measure at bring-up.